Lisp, 3^{rd} *Edition*

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Chapter 25 Streams and Delayed Evaluation

- Streams Are Sequences of Data Objects
- We Can Represent Streams Using Lists
- We Can Delay Evaluation by Encapsulation
- We Can Represent Streams Using Delayed Evaluation
- Problems
- Summary
- References

Chapter 26 Rule-Based Expert Systems and Forward Chaining

- Forward Chaining Means Working from Antecedents to Consequents
- We Use Streams To Represent Assertions and Rules
- Our First Pass Concentrates on `MATCH` and the Binding Stream
- Our Second Pass Concentrates on the Procedures that Surround `MATCH`
- Simple Rules Help Identify Animals
- Rules Facilitate Question Answering and Probability Computing
- Our Forward-Chaining Program Illustrates Abstraction
- Summary
- References

Chapter 27 Backward Chaining and Prolog

- Our Backward Chainer Borrows Procedures from our Forward Chainer
- Backward Chaining Means Working from Consequents to Antecedents
- Our First Pass Concentrates on `MATCH`, `UNIFY`, and the Binding Stream
- Our Second Pass Concentrates on the Procedures that Surround `MATCH` and `UNIFY`
- Completing Our Backward-Chaining Program Involves a Few Auxiliary Procedures
- Simple Rules Help Identify Animals
- Problems
- Our Backward Chainer Implements a Language like Prolog
- Problems
- Our Backward-Chaining Program Illustrates Abstraction